

Recursive Projection and Numerical Stability in State-Space Filtering: Square-Root Formulations of the Kalman Filter

Project Proposal (Alternative Topic)

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1. Introduction and Motivation

The Kalman filter is the standard tool for estimating the evolving hidden state of a linear system observed through noise. It is used wherever a quantity of interest cannot be measured directly but leaves a noisy trace over time: navigation and tracking, signal extraction, econometric nowcasting, and the estimation of unobserved components in time series.

Its textbook derivation is elegant, but its textbook implementation is numerically fragile. The covariance update takes the form of a difference of positive semi-definite matrices, and in finite-precision arithmetic that subtraction can destroy the symmetry and positive-definiteness that the covariance is mathematically guaranteed to possess. Once the computed covariance ceases to be positive definite, the filter gain is meaningless and the estimates diverge, often catastrophically and without warning. This is not a hypothetical concern: it is the reason square-root and factored formulations were developed for aerospace applications in the 1960s and 1970s, and it remains the reason they are preferred in practice today.

This project studies that gap between the mathematics and the arithmetic. It derives the filter as a recursive orthogonal projection, characterises the conditions under which the conventional recursion breaks down, implements the principal numerically stable reformulations, and compares them systematically on problems of controlled difficulty.

2. Mathematical Background

Consider the linear Gaussian state-space model

$$x(t) = F x(t-1) + w(t), \quad y(t) = H x(t) + v(t),$$

where $x(t)$ is the unobserved state, $y(t)$ the observation, and $w(t)$ and $v(t)$ are independent zero-mean noises with covariances Q and R respectively. The filtering problem is to compute the minimum mean-square-error estimate of $x(t)$ given observations up to time t .

The key structural fact is that this estimate is an orthogonal projection. In the Hilbert space of square-integrable random variables with inner product given by expectation of the product, the best linear estimate is the projection of the state onto the closed linear span of the observations,

and the estimation error is orthogonal to that span. The Kalman recursion is precisely the statement that this projection can be updated sequentially: each new observation contributes only through its innovation, the component orthogonal to everything already observed. This is the same projection-theoretic framework that underlies linear prediction and the innovations algorithm for stationary time series, specialised to a Markovian state model.

The error covariance obeys the Riccati recursion, whose measurement-update step is conventionally written

$$P = (I - K H) P^-,$$

with K the Kalman gain. Mathematically P is symmetric positive semi-definite. Numerically, the expression is a difference of two matrices of comparable magnitude, and is therefore subject to cancellation: computed eigenvalues can become negative when the problem is ill-conditioned, when the observation noise R is small relative to the state uncertainty, or when the state dimension is large. The Joseph form rewrites the same quantity as a sum of two positive semi-definite terms and is markedly more robust. Square-root filters go further, propagating a factor S satisfying $P = S S^T$ rather than P itself, so that positive semi-definiteness holds by construction and the effective condition number of the propagated object is the square root of that of P .

3. Research Questions

RQ1. Under what conditions, expressed in terms of the condition number of the error covariance, the ratio of process to observation noise, and the state dimension, does the conventional covariance recursion lose positive-definiteness and diverge in double-precision arithmetic?

RQ2. How do the Joseph-form update, the Cholesky-based square-root filter, and the UD (Bierman-Thornton) factorisation compare in numerical accuracy, robustness and computational cost across those regimes?

RQ3. Does the choice of formulation materially affect maximum-likelihood estimation of unknown model parameters, where the likelihood is evaluated through the filter's prediction-error decomposition and errors therefore propagate into the optimisation?

4. Methodology

Step 1. Derivation

Derive the filter from the projection theorem rather than by assuming Gaussianity, establishing the recursion as the sequential update of an orthogonal projection and identifying the innovation sequence as an orthogonal basis. Establish the Riccati recursion and state the conditions under which the error covariance converges to a steady state.

Step 2. Implementation

Implement four variants from first principles: the conventional recursion; the Joseph-form update; a square-root covariance filter based on Cholesky factors with QR-based propagation; and the UD factorisation. Verify all four agree to machine precision on well-conditioned

problems before proceeding, since agreement in the benign case is the correctness test.

Step 3. Controlled degradation

Construct simulated systems whose difficulty can be tuned continuously: shrink R towards zero to force near-perfect observations, increase the state dimension, and introduce near-unobservable modes so that the covariance becomes ill-conditioned in a controlled way. Because the data are simulated, the true state is known exactly, so estimation error is measured directly rather than inferred. Record, as a function of problem difficulty, the smallest eigenvalue of the computed covariance, the departure from symmetry, and the divergence of the state estimate from truth.

Step 4. Parameter estimation

Treat elements of F , Q and R as unknown and estimate them by maximising the Gaussian likelihood obtained from the prediction-error decomposition, using numerical optimisation. Compare the resulting estimates, and the conditioning of the optimisation itself, across filter formulations.

Step 5. Demonstration on real data

Apply the filter to one real univariate series in a local-level or local-linear-trend specification, to demonstrate signal extraction and to confirm that the behaviour observed under simulation is visible in practice. This step is deliberately positioned last: the methodological contribution does not depend on it.

5. Data and Computational Requirements

The methodological core of this project uses simulated data generated by the model itself, which means the true state is known and the project carries essentially no data-acquisition risk. The single real series required for Step 5 may be taken from any standard public repository; no restricted, purchased or licensed data is involved. All computation is performed in Python or R on a personal laptop, using standard numerical linear algebra routines, and requires no specialised hardware, cluster access or paid software.

6. Relationship to Prior Coursework

The project draws on three units directly. From Time Series and Random Processes come the projection theorem, linear prediction, the innovations algorithm and the state-space formulation itself. From Computational Linear Algebra come Cholesky and QR factorisation, matrix conditioning, and the analysis of finite-precision error propagation, which supply the tools for both the diagnosis and the remedy. From Optimisation and Operations Research comes the numerical maximisation of the likelihood surface in Step 4.

7. Expected Contribution

The individual algorithms studied here are established rather than novel. The contribution of the project is a systematic, self-contained and reproducible comparison, framed around the question of when the conventional formulation may safely be used and when it may not, a question of practical importance that is frequently answered by folklore rather than by measurement. The

work also serves as a worked demonstration that a mathematically exact recursion and a reliable numerical algorithm are not the same object, which is a point of general methodological value.

8. Limitations

- The analysis is confined to the linear Gaussian model. Nonlinear extensions such as the extended and unscented filters raise separate approximation-theoretic questions and are outside the intended scope.
- Conclusions about numerical behaviour are specific to IEEE double-precision arithmetic and to the particular linear algebra libraries used; they are empirical findings supported by theory, not hardware-independent theorems.
- The algorithms compared are established in the literature. The novelty lies in the systematic comparison and its framing, not in the derivation of new methods.

Indicative References

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